Half-life Answer Sheet

1. A sample of strontium-90 has a mass of 64 g. Its half-life is 29.1 years.

a) How long will it take for the sample to decay to 16 g?

Solution:

It will take 58.2 years.

Explanation: After one half life (29.1 years) there will be 32 g left.

After two half lives (29.1 x 2) there will be 16 g left.

b) How much strontium-90 will be left after 116.4 years?

Solution:

There will be 4 g left.

Explanation:

116.4 years is 4 half-lives. (116.4 ÷ 29.1 = 4)

After 1 half-life there is 32 g left. After 2 half-lives there is 16 g left. After 3 half-lives there is 8 g left. After 4 half-lives there is 4 g left.

2. An old animal bone sample contained 4.0×10^{-6} g of carbon-14 when the animal died. The bone now contains 2.5×10^{-7} g of carbon-14. How long ago did the animal die if carbon-14 has a half-life of 5730 years?

Solution:

22920 years ago.

Explanation:

After 1 half-life there will be 2×10^{-6} g left. After 2 half-lives there will be 1.0×10^{-6} g left. After 3 half-lives there will be 0.5×10^{-6} g left. Another way to write this is 5.0×10^{-7} g. After 4 half-lives there will be 0.25×10^{-7} g left.

4 half-lives of carbon-14 is 22920 years (5730 x 4).

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3. Answer TRUE if carbon dating is a good way to determine the age of the object, FALSE if it is not.

a. An old piece of parchment.

Answer: True. Parchment is an organic material, so it contains carbon. Carbon dating should be able to determine its age.

b. An ancient dinosaur fossil (older than 50,000 years).

Answer: False. A dinosaur fossil is organic material, so it contains carbon. It would be older than about 50,000 years old, however, so carbon dating would not work. After 50 000 years the amount of carbon-14 remaining in the fossil would be too small to detect.

c. An old metal knife.

Answer: False. A metal knife is not made of organic material, so its composition does not reflect the carbon in the atmosphere when it was made.

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